MIL-N-21945A(OS)
21 February 1973
SUPERSEDING
MIL-N-21945(Wep)
27 February 1959

MILITARY SPECIFICATION

Nonmagnetic Requirements For Parts for Mines Mark 56 and Mark 57

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification establishes the nonmagnetic requirements for procurement of parts (see 2.1) for the Mines Mark 56 and Mark 57.

2. APPLICABLE DOCUMENTS

2.1 The following standards, drawings and publications, or such portions thereof as designated herein of the issue in effect on the date of invitation for bids, form a part of this specification.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-481	Configuration Control-Engineering Changes, Deviations and Waivers (Short Form)

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

Dwg. 1186570	Ring
Dwg. 1186571	Shackle
Dwg. 1191746	Stop
Dwg. 1246626	Cover, Loading Hole
Dwg. 1274625	Strap
Dwg. 1299387	Spacer
Dwg. 1378573	Ring, Clamp
Dwg. 1379182	Magnetometer Housing Mark 1 Mod 0 Assembly
Dwg. 1386214	Spacer

FSC 1350

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Dwg. 1387843	Spacer
Dwg. 1387988	Strap
Dwg. 1418167	Spacer
Dwg. 1418168	Spacer
Dwg. 1553572	Shear Tube
Dwg. 1553573	Cage Weldment
Dwg. 1619172	Nut
Dwg. 1620227	Fuze
Dwg. 1620253	Weight
Dwg. 1620254	Weight
Dwg. 1620660	Nut
Dwg. 1634067	Nut, Hex, Machine Screw
Dwg. 1634068	Washer, Flat
Dwg. 1635006	Shear Tube
Dwg. 1700974	Lockwasher
Dwg. 1700998	Washer
Dwg. 1738313	Cap, Screw, Socket Head
Dwg. 1738318	Screw, Slotted, Pan Head
Dwg. 1825736	Coupling Nut
Dwg. 1826627	Screw, Slotted Pan Head
Dwg. 1826628	Screw, Machine, Flat Head
Dwg. 1826629	Retaining Ring, E Type
Dwg. 1826630	Retaining Ring
Dwg. 1827674	Nut, Hex, Self Locking
Dwg. 1827675	Screw, Cap
Dwg. 1827687	Bolt, Hex Head
Dwg. 2155262	Spacer
Dwg. 2280626	Spacer
Dwg. 2289340	Spring Pin
Dwg. 2420946	Filter, Radio Interference, Mark 4 Mod 0
Dwg. 2422431	Filter, Radio Interference, Mark 3 Mod 0
Dwg. 2422446	Capacitor, Fixed, Mylar-Dielectric
Dwg. 2423304	Support
Dwg. 2426520	Retainer Assembly
Dwg. 2491852	Arming Wire Mark 5 Mod 1

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

OD 10511	Basic Methods for Nonmagnetic Testing with the Test Set Mark 115 Type

(Copies of standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Description. This specification lists those parts for the Mines Mark 56 and 57 which shall be subjected to magnetic effects tests when procured separately.

3.2 General requirements.

3.2.1 Requests for deviation. Requests for deviation from this specification, applicable drawings, standards, publications, materials and processes specified herein, shall be as set forth in MIL-STD-481.

3.2.2 Conflicting requirements. Conflicting requirements arising between this specification or any standard, publication or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Performance requirements and product characteristics.

3.3.1 Static magnetic effect. When tested in accordance with 4.5.2, the static magnetic effect of the parts as tested shall not exceed 0.10 milligauss when measured as described in OD 10511.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection and testing of the spare and special parts shall be classified as follows:

a. First article inspection (paragraph 4.5)

b. Quality conformance inspection (paragraph 4.5)

c. Periodic production sample inspection (paragraph 4.5)

4.3 Inspection.

4.3.1 First article inspection. The first article sample of the item being procured shall be tested by the contractor, or an

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activity designated by the procuring agency, to determine that it meets the requirements of this specification.

4.3.2 Quality conformance.

4.3.2.1 Sampling. Acceptance samples selected from the production lot for testing by the contractor or a Government-designated testing activity shall be randomly selected in accordance with MIL-STD-105, Inspection Level II, AQL 1.0 percent or as designated on the specification or drawing for the particular part. In the event of a conflict between the AQL specified herein and the specification or drawing for the particular part, the specification or drawing shall have precedence.

4.3.2.2 100 percent inspection. Unless otherwise specified, inspection shall be performed 100 percent only on those items where 100 percent inspection is required by the applicable specification or drawing.

4.3.3 Periodic production sample test. When specified, a periodic production sample of the applicable item shall be randomly selected from the production lot for submission to an activity designated by the procuring agency to determine if the item meets the requirements of this specification. Nonperformance of this test by the contractor does not relieve the contractor from the responsibility of meeting the requirements of this specification.

4.4 Test equipment. The following item of test equipment is required to perform the test of 4.5:

Test Set Mark 115 (Government furnished)

4.5 Tests.

4.5.1 General test conditions and definitions. The general test conditions and definitions shall be as specified in OD 10511.

4.5.2 Test procedure. Static magnetic effect tests shall be accomplished as specified in OD 10511, Test Method I. For the purpose of accomplishing the tests, the test distance and the rate of rotation shall be as specified in Table I or II for each category of parts to be tested. Note that the rate of rotation must be slow for large aluminum or copper parts, especially if these parts are cylindrical or spherical, to prevent the production of magnetic eddy current effects. A smooth, continuous rotation is required for such parts; avoid abrupt motions.

4.5.2.1 Small parts. Small parts to be tested shall be arranged in a compact manner in a volume whose base area shall not exceed 3 inches square and whose height shall not exceed 1-1/2 inches.

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The parts shall be secured in this volume to insure no relative movement during testing. In order to meet these conditions, the parts can be tightly packed in a suitable nonmagnetic container or can be secured together with cord or tape. Table I specifies the minimum quantity of parts that shall be tested at one time. If the quantity on hand is less than the minimum specified, approval must be obtained from the procuring agency before beginning nonmagnetic tests. At the discretion of the testing activity, a quantity greater than the minimum specified may be tested at one time; however, the static magnetic effect of such a quantity shall not exceed 0.10 milligauss.

4.5.2.1.1 The test face for the small parts to be tested shall be the base area of the grouped parts. The test distance shall be 1-3/4 ± 1/16 inches from the center of the test magnetometer to the test face. A convenient method of maintaining this test distance during testing is by the use of a 1/4-inch thick, flat, nonmagnetic spacer 3 inches square secured either to the appropriate faces of the magnetometer or to the base area of the grouped parts. The test axis shall be the line perpendicular to and passing through the center of the base area of the grouped parts. In testing, the test face shall be held at the test distance in each of the three orientations specified in OD 10511.

4.5.2.2 Large parts. Table II specifies the minimum quantity that must be tested and the maximum that may be tested at the discretion of the testing activity. When more than a single item is tested, the parts shall be secured together to prevent shifting or movement of the parts during the test.

4.5.2.2.1 The test face for the large parts to be tested shall be either one of the larger area faces of the test material. The test distance from the magnetometer center to the test face is specified in Table II. The test axis shall be perpendicular to the test face and shall pass through the center of mass of the part. In testing, the test face shall be at the test distance from the test magnetometer in each of the three orientations in OD 10511.

4.5.2.3 Magnetometer Housing Mark 1 Mod 0. The magnetic testing of the Magnetometer Housing Mark 1 Mod 0 shall be conducted in accordance with OD 10511, Test Method 1, with each of the six sides of the magnetometer housing considered as a test face and tested accordingly. The test distance from the center of the magnetometer to the face being tested shall be 1-3/4 ± 1/16 inches. The four faces of the magnetometer housing perpendicular to the mounting flange shall be presented to the test magnetometer by placing the extending edge of the mounting flange at the prescribed test distance with the face under test parallel to the adjacent test magnetometer face. The magnetometer housing shall be substantially centered with respect to the test magnetometer during each of the 18 test presentations. The rate of rotation for Orientation C shall be 8 to 15 seconds per revolution.

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4.5.2.4 Arming Wire, Mark 5 Mod 1. The magnetic testing of the Arming Wire Mark 5 Mod 1 shall be conducted in three test cycles as shown in 4.5.2.4.1, disregarding provisions of 4.5.2. Each test cycle shall consist of the following parts:

(1) Calibrate signals "+" and "-" of suitable magnitude. Indicate sign and value of signals on the recorded tapes.

(2) Hold the test object at the reference position (any position sufficiently removed from the magnetometer where the magnetic signal is less than 0.01 milligauss) for a minimum of three seconds.

(3) Bring the test object to the test position (see 4.5.2.4.1) and move as directed.

(4) Remove the test object to the reference position and hold for a minimum of three seconds.

(5) Calibrate signal "+" and "-" of suitable magnitude. Indicate value of signals on the recorded tapes.

4.5.2.4.1 Motion for test cycles. The motion required for each of the three test cycles is described as follows:

Motion for Test Cycle 1. Beginning at one end of the inner packing container which holds 50 arming wires, the package shall be moved longitudinally and tangentially along the face of the magnetometer marked "TOP" at a speed of 2 to 6 inches per second. (Under these conditions, the metal will not come closer to the face of the magnetometer than 1/2 inch due to the thickness of the packaging material.)

Motion for Test Cycle 2. With the container held parallel to the background field and beginning at one end, the package shall be passed along the side of the magnetometer for approximately one half of its length at the same speed as previously mentioned. (Because of the length of the package, only one half can be conveniently tested at a time.)

Motion for Test Cycle 3. By turning the package endwise, the motion for Test Cycle 2 may be continued to test the entire length of the package. (Note. In the event that a reading greater than 0.10 milligauss is obtained, the bundle of long sections should be removed from the inner container and retested while maintaining the same spacing distance, 1/2 inch from the magnetometer, since magnetic test is not specifically required on the short sections.)

4.5.2.4.2 A reading in excess of 0.10 milligauss peak-to-peak for any test cycle except as explained in note above shall be cause for rejection.

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4.5.3 Acceptance criteria. Any part which fails to meet the product characteristics requirement of this specification shall be regarded as defective.

5. PREPARATION FOR DELIVERY (Not applicable)

6. NOTES

6.1 Intended use. This specification establishes the magnetic effects test requirements for nonmagnetic parts used in the Mines Mark 56 and 57 when tested with the Test Set Mark 115 in accordance with OD 10511.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Exceptions to this specification.
c. Applicable drawings and other documents.

6.2.2 Contract data requirements. Test reports associated with any of the tests covered in paragraph 4.5 of this specification will be prepared and delivered by the contractor only when they are listed on a Contract Data Requirements List (DD Form 1423) incorporated into the contract. In those instances, the reports shall be prepared in accordance with the following Data Item Description (DID) cited by number in Block 4 of the DD Form 1423:

DID No.	Title
DI-T-2072	Report, Test

Preparing Activity

Navy - OS

Project No.

1350-N027

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TABLE I

Drawing	Name	Min. Test Quantity	Rate of Rotation in Sec./Rev.
1386214	Spacer	6	5 to 15
1619172	Nut	2	5 to 15
1620227	Fuze	4	5 to 15
1620660	Nut	8	10 to 25
1634067-11	Hex Nut	12	5 to 15
1634067-12	Hex Nut	18	5 to 15
1634067-13	Hex Nut	9	5 to 15
1634067-14	Hex Nut	6	5 to 15
1634067-15	Hex Nut	6	5 to 15
1634068-7	Washer	48	5 to 15
1634068-12	Flat Washer	45	5 to 15
1634068-14	Flat Washer	18	5 to 15
1634068-18	Flat Washer	8	5 to 15
1700974-4	Lock Washer	48	5 to 15
1700974-5	Lock Washer	48	5 to 15
1700974-6	Lock Washer	48	5 to 15
1700974-7	Lock Washer	48	5 to 15
1700974-8	Lock Washer	18	5 to 15
1700974-9	Lock Washer	18	5 to 15
1700974-10	Lock Washer	12	5 to 15
1700998	Washer	12	5 to 15
1738313-23	Soc. Hd. Cap Screw	8	5 to 15
1738313-34	Soc. Hd. Cap Screw	8	5 to 15
1738313-46	Soc. Hd. Cap Screw	8	5 to 15
1738313-47	Soc. Hd. Cap Screw	8	5 to 15
1738313-52	Soc. Hd. Cap Screw	4	5 to 15
1738313-53	Soc. Hd. Cap Screw	4	5 to 15
1738313-56	Soc. Hd. Cap Screw	8	5 to 15
1738313-62	Soc. Hd. Cap Screw	4	5 to 15
1738313-64	Soc. Hd. Cap Screw	4	5 to 15
1738313-77	Soc. Hd. Cap Screw	2	5 to 15
1738313-97	Soc. Hd. Cap Screw	4	5 to 15
1738318-61	Screw, Slotted, Pan Hd.	24	5 to 15
1738318-62	Screw, Slotted, Pan Hd.	24	5 to 15
1739318-64	Screw, Slotted, Pan Hd.	16	5 to 15
1825736	Coupling Nut	1	5 to 15
1826627-77	Screw, Sl. Pan Hd.	18	5 to 15
1826627-79	Screw, Sl. Pan Hd.	12	5 to 15
1826628-68	Screw, Mach. Fl. Hd.	24	5 to 15
1826629-157	Retaining Ring	24	5 to 15
1826630-77	Retaining Ring	12	5 to 15
1827674-292	Nut, Hex, Self-Locking	6	5 to 15

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TABLE I (Continued)

1827674-439	Nut, Hex, Self-Locking	12	5 to 15
1827674-463	Nut, Hex, Self-Locking	6	5 to 15
1826675-93	Cap Screw	2	5 to 15
1827687-1	Bolt, Hex Head	6	5 to 15
2155262-1 & 2	Spacer	6	5 to 15
2422446	Capacitor, Fixed	12	5 to 15
2289340-2	Pin, Spring	4	5 to 15
2289340-4	Pin, Spring	4	5 to 15

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TABLE II

Drawing	Test Distance	Test Quantity Min.	Test Quantity Max.	Rate of Rotation Sec/Rev
1186570	1-3/4 ± 1/16	1	2	5 to 15
1186571	1-3/4 ± 1/16	1	2	5 to 15
1191746	1-3/4 ± 1/16	1	4	5 to 15
1246626	1-3/4 ± 1/16	1	3	5 to 15
1274625	2-1/2 ± 1/16	1	4	5 to 15
1299387	1-3/4 ± 1/16	10	25	5 to 15
1378573	5 ± 1/8	1	1	5 to 15
1379182	See para. 4.5.2.3	1	1	See para. 4.5.2.3
1387843	1-3/4 ± 1/16	1	1	5 to 15
1387988	2-1/2 ± 1/16	1	4	5 to 15
1418167-8	1-3/4 ± 1/16	10	25	5 to 15
1418168	1-3/4 ± 1/16	10	25	5 to 15
1553572	2-1/2 ± 1/16	1	1	5 to 15
1553573	1-3/4 ± 1/16	1	1	10 to 15
1620253	2-1/2 ± 1/16	1	1	5 to 15
1620254	2-1/2 ± 1/16	1	1	5 to 15
1635006	2-1/2 ± 1/16	1	1	5 to 15
1827687-2	1-3/4 ± 1/16	2	6	5 to 15
2280626	1-3/4 ± 1/16	1	1	5 to 15
2420946	7 ± 1/8	1	1	5 to 15
2422431	7 ± 1/8	1	1	5 to 15
2423304	2-1/2 ± 1/16	1	2	5 to 15
2426520	2-1/2 ± 1/16	1	2	5 to 15
2491852	See para. 4.5.2.4			

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